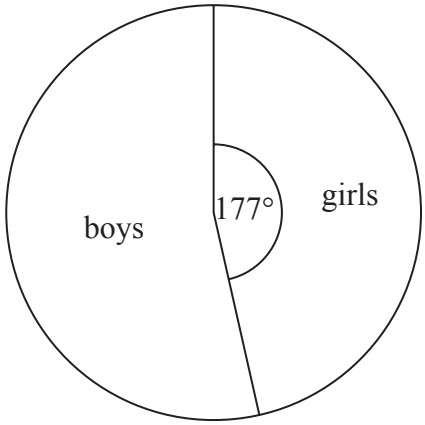
21 There are 240 students in Year 7 at a school.

The pie chart shows the proportion of boys and the proportion of girls in Year 7



There are 8 more girls in Year 8 than in Year 7
There are 32 fewer boys in Year 8 than in Year 7

Andy draws a pie chart to show the proportion of boys and the proportion of girls in Year 8

Work out the angle of the sector in Andy's pie chart that represents girls.

Year 7: 177° out of 360° represents the girls.

	Deg	Pupils	
	360	240	
$\times \frac{59}{120}$			$\times \frac{59}{120}$
	177	118	

Girls in Year 7 = 118.

1 mark
177/360 of 240 = 118

Year 8 girls = 118 + 8 = 126.

1 mark
126 girls; 216 total

Year 8 total = 240 + 8 − 32 = 216.

	Deg	Pupils	
	360	216	
$\times \frac{7}{12}$			$\times \frac{7}{12}$
	210	126	

Angle = 126/216 × 360°

= 210°.

1 mark
126/216 of 360°

210 °

1 mark
210°

(Total for Question 21 is 4 marks)